

Civics Group	Index Number	Name (use BLOCK LETTERS)	H2
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ST. ANDREW'S JUNIOR COLLEGE
2021 JC2 PRELIMINARY EXAMINATIONS

H2 BIOLOGY

Paper 2

9744/2

Monday

30th August 2021

2 hours

Materials: Question Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
Write your answers in the spaces provided on the question paper.

The number of marks is given in brackets [] at the end of each question or part question.

				For Examiners' Use	
				1	/10
				2	/12
				3	/10
				4	/9
				5	/6
				6	/11
				7	/12
				8	/12
				9	/11
				10	/7
				Total	/100

Conceptual error (C)	Data Quoting (D)	Expression (E)	Misreading the question (Q)

This document consists of **27** printed pages.

[Turn over]

Question 1

(a) T-helper lymphocytes and Leydig cells are two types of mammalian cells. The main role of T-helper lymphocytes and Leydig cells is to synthesise and secrete cell-signalling molecules.

- T-helper lymphocytes synthesise proteins known as cytokines.
- Leydig cells synthesise the steroid (lipid) hormone testosterone from cholesterol. Leydig cells also synthesise cholesterol.

Figure 1.1 shows part of a mammalian cell.

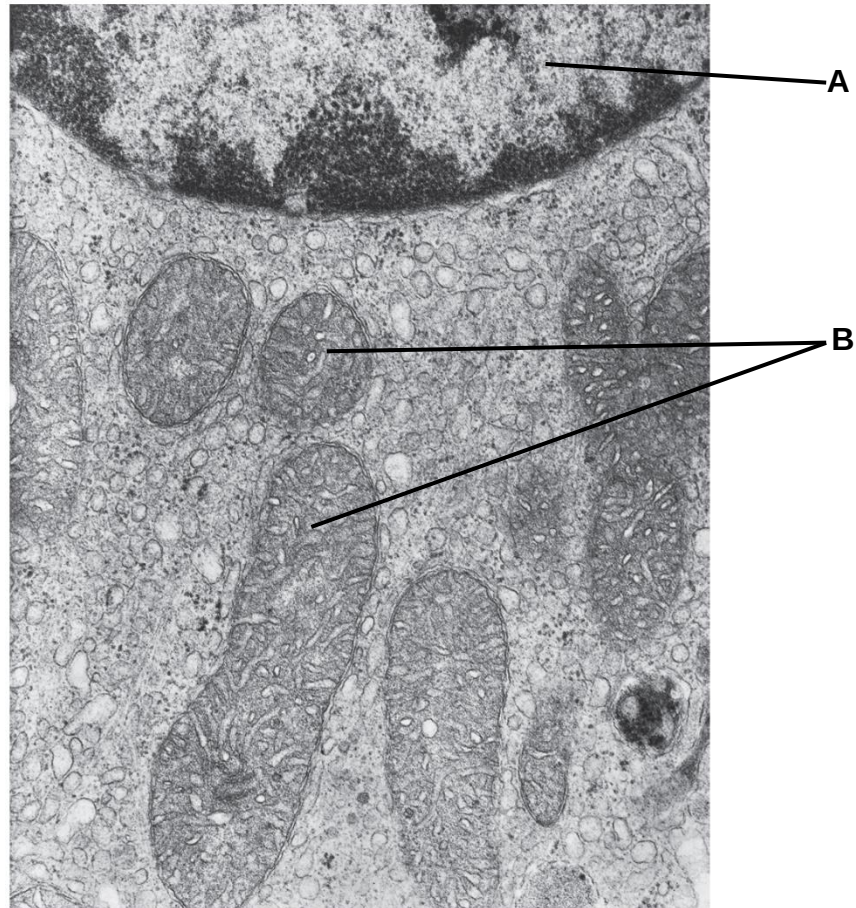


Fig. 1.1

(i) Identify organelles **A** and **B** on Fig. 1.1.

A:

B:

[1]

- (ii) Organelle **A** is involved in mitotic cell cycle. State the process that occurs during interphase to produce genetically identical daughter cells.

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.....[1]

- (iii) Explain whether Fig. 1.2 shows part of a Leydig cell or part of a T-helper lymphocyte.

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.....[2]

- (b) Cell-surface receptors are membrane-anchored (integral) proteins that bind to external signalling molecules.

Such receptors are synthesised within the cell and brought to the cell surface membrane as shown in the Figure 1.2.

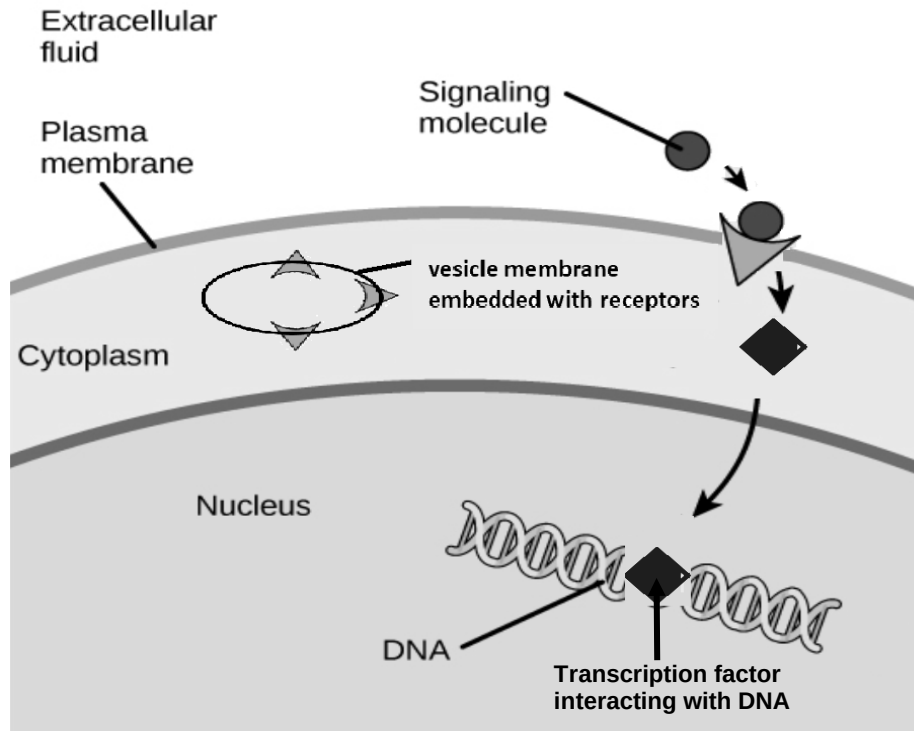


Fig. 1.2

With reference to Fig. 1.2, describe how the receptor is brought to and become part of the cell surface membrane.

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.....[2]

- (c) *Eichhornia crassipes*, a water hyacinth plant, is an invasive species found in lakes and rivers throughout Asia. They grow rapidly, producing many flowers and seeds. Each flower of this plant has six anthers (each anther is at one end of the filament) and one stigma as shown in Figure 1.3.

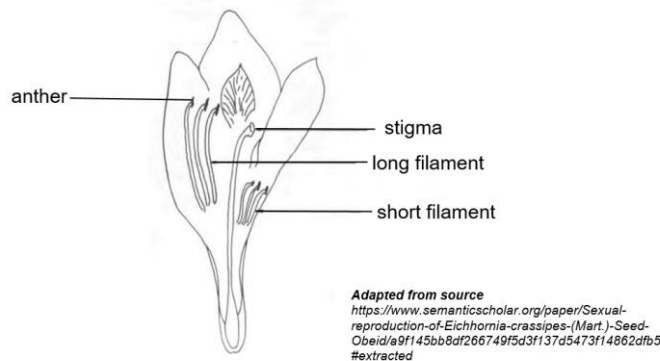


Fig. 1.3 showing spatial arrangement of stamen (comprising of both anther and filament) and stigma in *E. crassipes*.

Each anther was observed to release many pollen grains. After pollination, the pollen grains germinate after it comes into contact with the stigma, in response to the sugary fluid secreted by the mature stigma. A pollen tube grows out from each pollen grain. The pollen tube serves to transport the male gamete cells to the female gamete cells (ovules) in the flower.

The tip of the pollen tube is protected by an outer layer of pectin.

Pectin is a polysaccharide.

Starch grains, mitochondria and vesicles are present in the tip of the pollen tube.

Using the information given and your content knowledge on organelles and carbohydrates, describe how the production of pectin occurred at the tip of a pollen tube.

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[Total: 10]

QUESTION 2

Rhodopsin is a cell signalling receptor found in the rod cells of the retina that is associated with retinal thinning and degeneration. There have been correlations established between retinal thinning and Alzheimer's disease. Hence, scientists have realised the potential of the study of rhodopsin in the early detection and progression monitoring of Alzheimer's disease.

Figure 2.1 shows the signalling pathway of rhodopsin in rod bipolar cells.

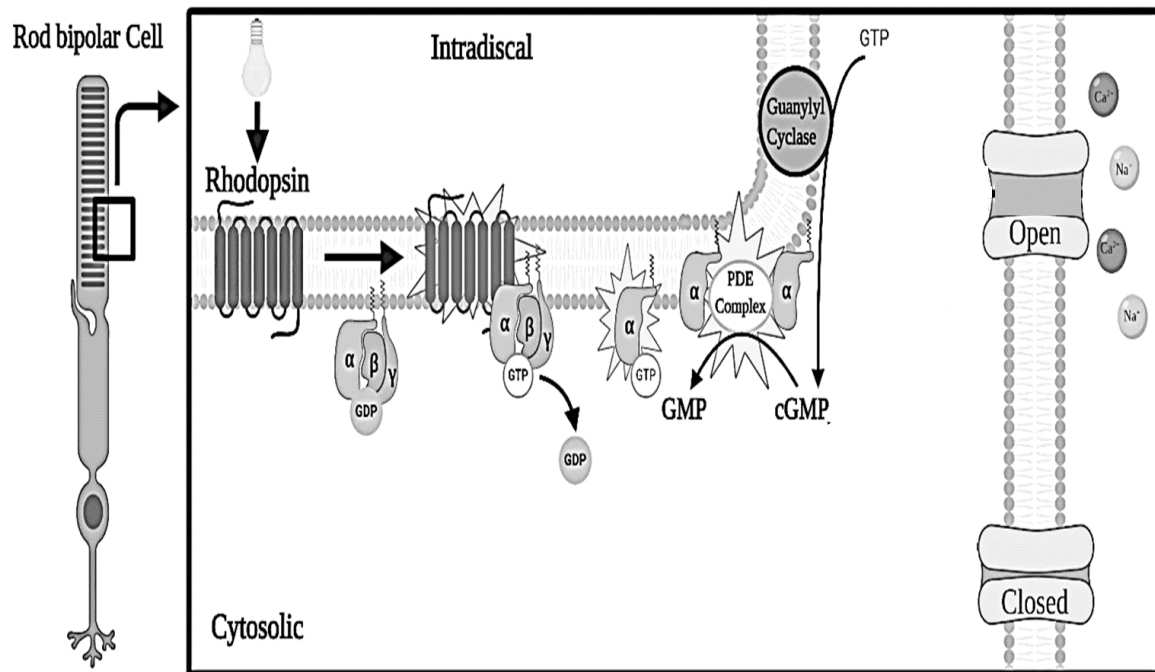


Fig. 2.1

(a)(i) With reference to Fig. 2.1, state the class of receptors in which rhodopsin belongs to. Justify your answer.

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.....[2]

(ii) Explain how the structure of rhodopsin relates to its function.

.....[4]

When light activates rhodopsin, it initiates a sequence of events that consequently decreases the concentration of cGMP. When the cGMP concentration decreases, the cGMP-gated channels will close, preventing depolarization of the membrane induced by the influx of Na^+ and Ca^{2+} .

(b) With reference to Fig. 2.1, describe how activation of rhodopsin can lead to a decrease in cGMP concentration.

.....[5]

Over-activation of rhodopsin from light results in prolonged depolarization in membrane potential. This could accelerate retinal thinning and degeneration. As such, over the years, cells have evolved to develop a mechanism to prevent the over-activation of rhodopsin.

(c) Suggest one mechanism that the cell can use to prevent the over-activation of rhodopsin.

.....[1]

[Total: 12]

QUESTION 3

ATP is often known as the universal 'energy currency' of the cell.

Most ATP are made in cells by membrane systems that create proton gradients by pumping protons from one compartment to another.

Figure 3.1 shows two of such membrane systems.

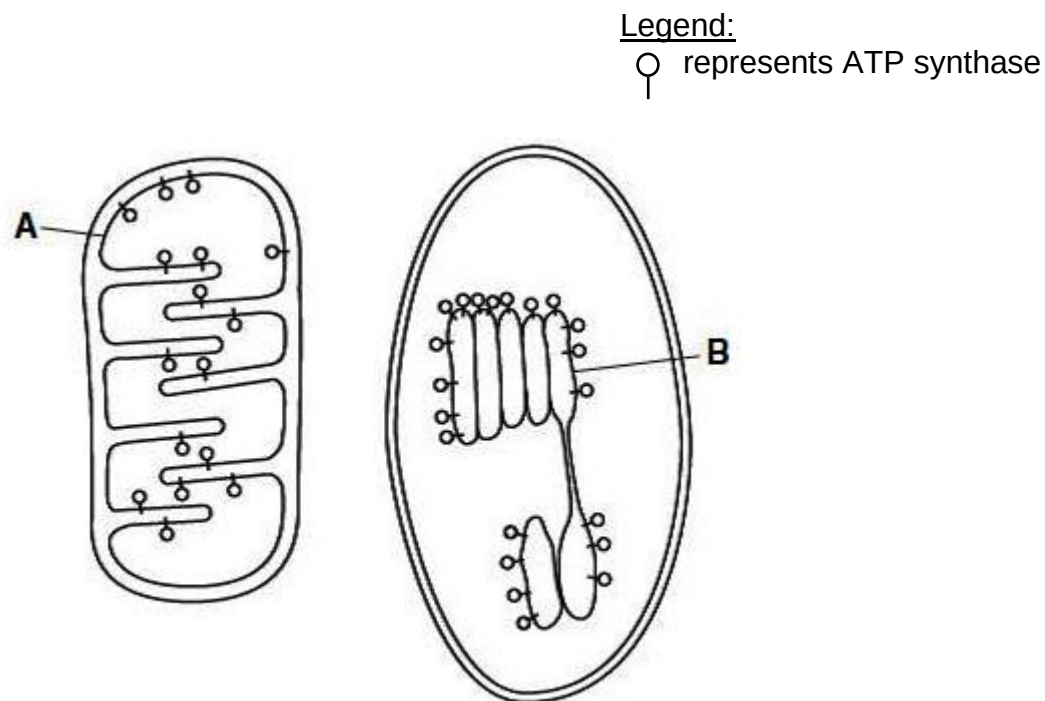


Fig. 3.1

(a)(i) Draw arrows onto **each** of the membrane systems in Fig. 3.1 to show the direction in which protons are pumped.

.....[1]

(ii) Outline how energy is generated for the pumping of protons across membrane system B.

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(iii) Explain why transmembrane proteins are necessary for protons to be pumped across the membrane systems.

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It is also known that there are cyclic pathways involving the use of enzymes occurring in both membrane systems.

(b) State the names of the two cyclic pathways and outline the differences between them.

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[Total: 10]

Question 4

(a) Figure 4.1 shows the nematode worm, *Caenorhabditis elegans*.



Fig. 4.1

Stem cells of *C. elegans* have been studied.

Figure 4.2 shows the change in mass of DNA per nucleus in a stem cell during one cell cycle. $1\text{pg} = 1 \times 10^{-12}\text{g}$.

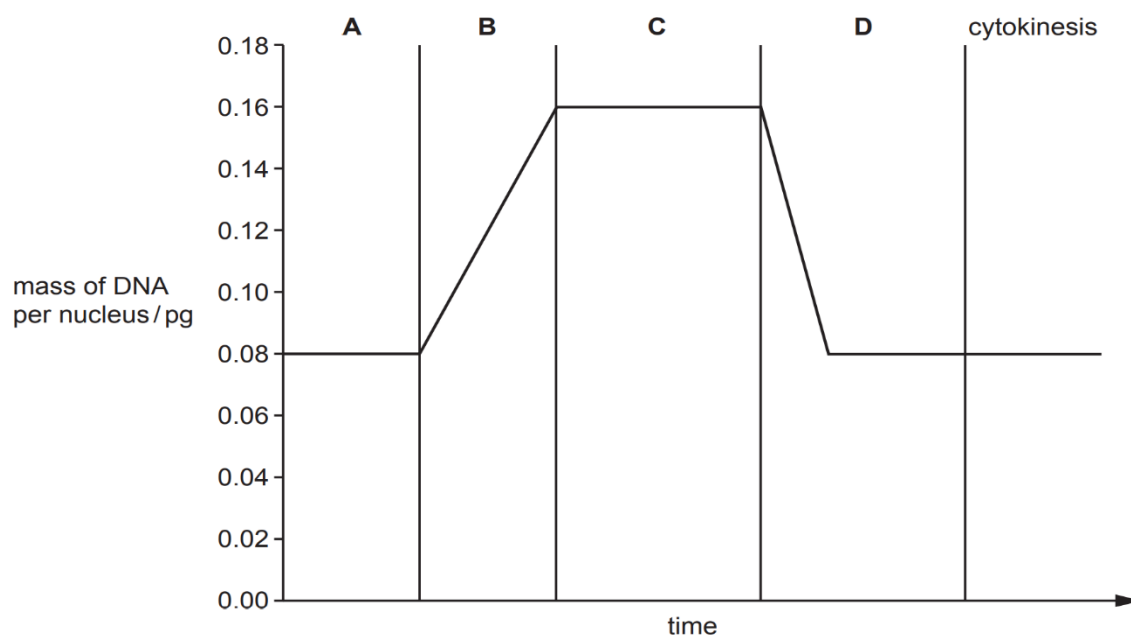


Fig. 4.2

- (i) State the phases of interphase shown by **A** and **B** and the stage of mitosis shown by **D** in Fig. 4.2.

A

B

D

[1]

(ii) Outline what happens in a cell in stage D in preparation for cytokinesis.

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.....[2]

(iii) A stem cell of a female *C. elegans* has 12 chromosomes.

Complete Table 4.1 to show the number of nuclei within the stem cell in stages **A**, **B** and **D** of the cell cycle shown in Figure 4.2 and the number of chromosomes in each nucleus during these stages.

Table 4.1

	stage of the cell cycle		
	A	B	D
number of nuclei within the stem cell			
number of chromosomes in each nucleus			

[2]

(iv) Immediately after cytokinesis, daughter cells are **not** identical even though they are genetically identical.

Suggest a reason why daughter cells are not identical immediately after cytokinesis.

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.....[1]

- (b) The tomato plant, *Solanum lycopersicum*, does not tolerate periods of drought (water shortage). Researchers have produced a tomato plant that has an improved tolerance of drought.

The nuclei of plants produce small lengths of RNA known as microRNAs.

MicroRNAs in guard cells have been shown to prevent the synthesis of some proteins.

The guard cells of the drought-tolerant tomato plants produced more microRNA molecules than the guard cells of the non-tolerant plants.

MicroRNA molecules do **not** prevent transcription but interact with messenger RNA (mRNA).

Suggest how this microRNA can interact with mRNA to prevent the production of proteins in guard cells of *S. lycopersicum*.

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[Total: 9]

QUESTION 5

RNA polymerase in *Mycobacterium tuberculosis* is composed of five different polypeptides. Gene *rpoB* codes for one of these polypeptides known as the β -subunit.

One or more mutations in a specific region of *rpoB* result in strains of *M. tuberculosis* that are resistant to rifampicin, an antibiotic. In these strains, mutations often occur in two DNA triplets in this region, in positions 526 and 531.

Table 5.1 summarises the results of an investigation into seven rifampicin-resistant strains, A to G, that have amino acid changes for positions 526 and 531.

Table 5.1 includes:

- the change in the **mRNA codon** for position 526 or position 531
- the amino acid change that has occurred as a result of the mutation
- the minimum concentration of rifampicin required to inhibit growth of the bacterial strain (MIC)
- the number of **other** mutations occurring within the specific region of *rpoB*.

Table 5.1

Key

$\approx$ approximately

$\geq$ greater than or equal to

$\leq$ less than or equal to

strain	codon involved	mRNA codon change	amino acid change	MIC / $\mu\text{g cm}^{-3}$	number of other mutations in the specific region
A	526	CAC $\rightarrow$ UAC	His $\rightarrow$ Tyr	≤ 50	0
B	526	CAC $\rightarrow$ AAC	His $\rightarrow$ Asn	≥ 100	1
C	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	$\approx 50-75$	2
D	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	≥ 100	3
E	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	≈ 50	3
F	526	CAC $\rightarrow$ UUC	His $\rightarrow$	≥ 100	3
	531	UCG $\rightarrow$ UUG	Ser $\rightarrow$ Leu		
G	526	CAC $\rightarrow$ UAC	His $\rightarrow$	≥ 100	3
	531	UCG $\rightarrow$ UUC	Ser $\rightarrow$ Phe		

(i) Complete Table 5.1 to show the amino acid changes that have occurred in strains **F** and **G**. [1]

(ii) With reference to Table 5.1, list the strains of *M. tuberculosis* that show the greatest resistance to rifampicin.

.....[1]

(iii) Suggest reasons to explain why strains **C** and **D** show:

- resistance to rifampicin
- different levels of resistances to rifampicin.

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[Total: 6]

QUESTION 6

Skin colour in humans is classified according to the Von Luschan's chromatic (colour) scale as seen in Table 6.1.

Table 6.1

Colour	Sunburning	Tanning behavior	Von Luschan's chromatic scale
Light, pale white	Always	Never	0–6
White, fair	Usually	Minimally	7–13
Medium white to light brown	Sometimes	Uniformly	14–20
Olive, moderate brown	Rarely	Easily	21–27
Brown, dark brown	Very rarely	Very easily	28–34
Very dark brown to black	Never	Never	35–36

(a) With reference to Table 6.1, state and explain the type of variation depicted by the trait of skin colour.

Type of variation:

Explanation:

.....[2]

(b) Fur colour in cats is also controlled by genes.

The gene coding for colour is located in the X chromosome. Allele O gives rise to orange fur, while allele B gives rise to black fur. Both allele O and allele B are co-dominant to each other. A cat with both alleles have a tortoise shell variegation.

Another gene E on an autosome (non-sex chromosome) affects the overall colour of the fur. One dominant allele of this gene (allele E) prevents normal development of melanocytes (pigment producing cells). Therefore, cats with genotype (E_) will have entirely white fur regardless of the genotype at the gene locus for colour.

(i) State the type of gene interaction between the two genes.

Type of interaction:

[1]

(ii) A female cat with tortoise shell variegation and a white male cat which is the offspring of a black female cat are chosen to mate.

Draw a genetic diagram to show the outcomes of this cross.

.....[4]

(c) A chi-square analysis was performed on another group of cats. There was a total of six phenotypic classes in the F1 generation.

chi-squared table:

Deg. of freedom	p= 0.5	p= 0.1	p= 0.05	p= 0.01	p= 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

The calculated chi-square value for the test is found to be 8.15.

(i) Use this information and make some valid conclusions for this analysis.

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(ii) Suggest a way to ensure that the data obtained from observing the results of the cross is statistically reliable.

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.....[1]

[Total: 11]

QUESTION 7

Each human cell contains approximately 2 meters of DNA if stretched end-to-end; yet the nucleus of a human cell, which contains the DNA, is only about 6 μm in diameter.

(a) Describe how DNA is packaged to fit into the nucleus of eukaryotic cells.

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.....[4]

(b) The ends of each DNA molecule are lengths of DNA called telomeres.

In humans, telomeres are made up of the base sequence TTAGGG, which is repeated up to 400 times.

In some cells, each time a DNA molecule is replicated, the last 150 - 200 bases are not copied. The telomeres therefore get shorter with each DNA replication until they become too short for DNA replication to take place.

Calculate the maximum number of replications that can occur with the DNA that contains 400 TTAGGG repeats.

Show your working.

Maximum number of replications = [2]

Eukaryotic DNA contains both introns and exons.

Figure 7.1 shows the structure of a length of pre-mRNA formed from one gene.

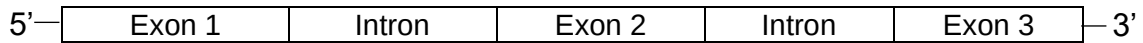


Fig. 7.1

(c) (i) Explain how the pre-mRNA molecule shown in **Fig. 7.1** is converted into a mature mRNA molecule.

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.....[3]

(ii) Sketch the longest mature mRNA molecule that can be formed from the pre-mRNA shown in Fig. 7.1.

.....[1]

The processing pre-mRNA molecule to a mature mRNA molecule does not occur in prokaryotes.

(iii) Explain two other differences in the control of gene expression between eukaryotes and prokaryotes.

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[Total: 12]

QUESTION 8

- (a) Coronavirus disease 2019 (COVID-19) is a pandemic affecting many countries globally. COVID-19 is caused by a novel coronavirus, severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Figure 8.1 shows the structure of SARS-CoV-2.

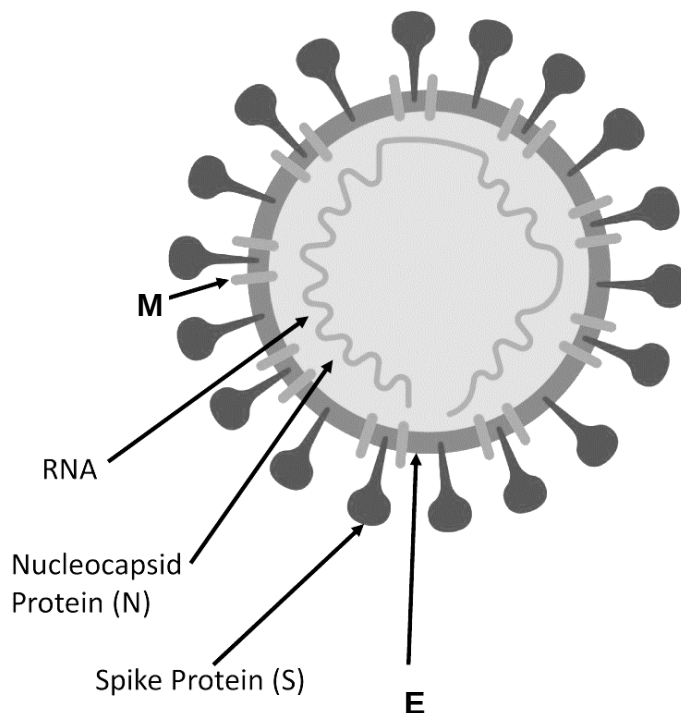


Fig. 8.1

- (i) Identify the monomer(s) of structure **M** and the main constituent of structure **E**.

monomer(s) of structure **M**:

main constituent of structure **E**:

[2]

Figure 8.2 shows a simplified version of the reproductive cycle of SARS-CoV-2.

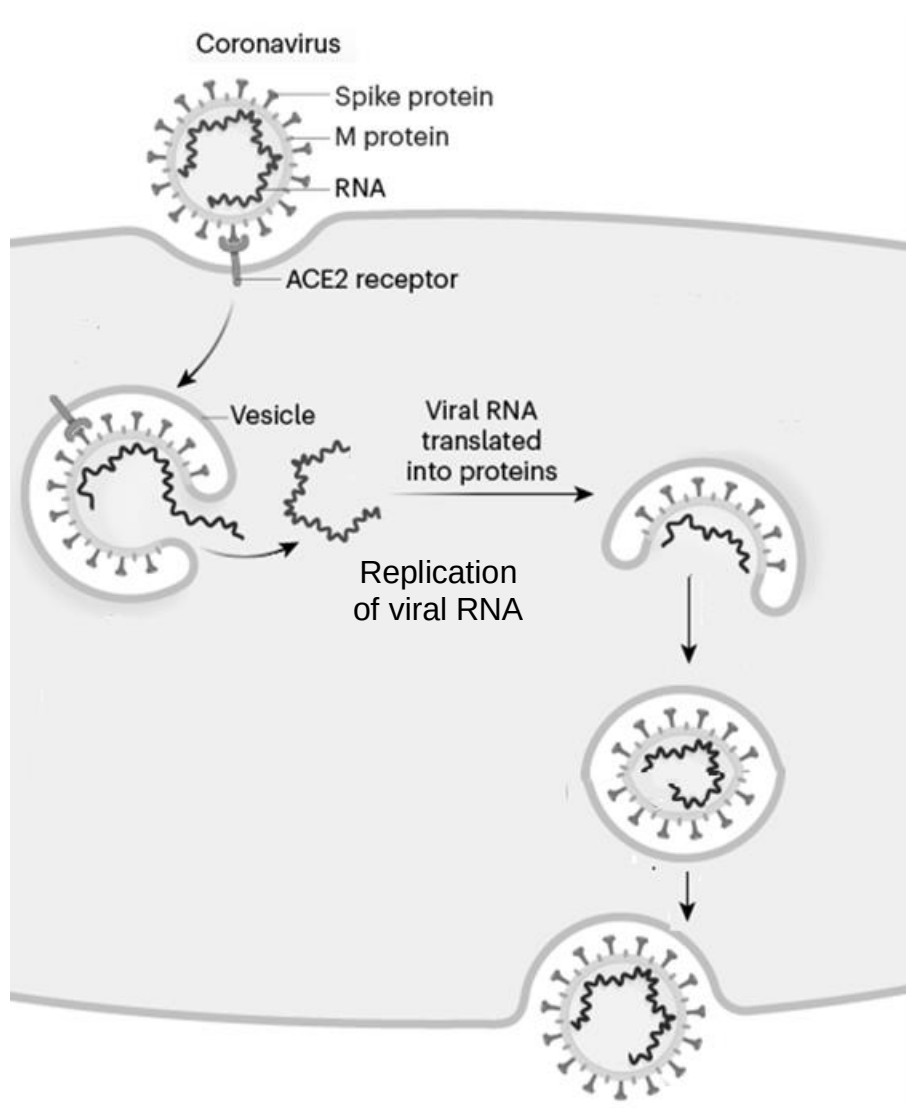


Fig. 8.2

(ii) With reference to Fig 8.2 and your biological knowledge, compare the reproductive cycle of SARS-CoV-2 with that of influenza virus.

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- (b) It was discovered that the SARS-CoV-2 genome was positive sense RNA and that its capsid also contained viral RNA dependent RNA polymerase.

Based **solely** on the information provided in this question, suggest how new variants of SARS-CoV-2 are formed.

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- (c) COVID-19 is commonly detected in individuals via swab tests, which involve the use of real time Reverse Transcription Polymerase Chain Reaction (RT-PCR). A sample is first collected from parts of the body where SARS-CoV-2 gathers, such as a person's nose or throat. The sample is then broken down to expose the genetic materials and RT-PCR is then performed to determine the viral load.

- (i) Explain why both reverse transcription and PCR are needed in order to detect the virus.

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.....[2]

- (ii) Four individuals, W X Y & Z were placed under quarantine order by the Ministry of health. After 13 days, they took the exit swab test (involving RT-PCR). Subsequently, gel electrophoresis was carried out on the respective RT-PCR products synthesised and the result is shown in Figure 8.3.

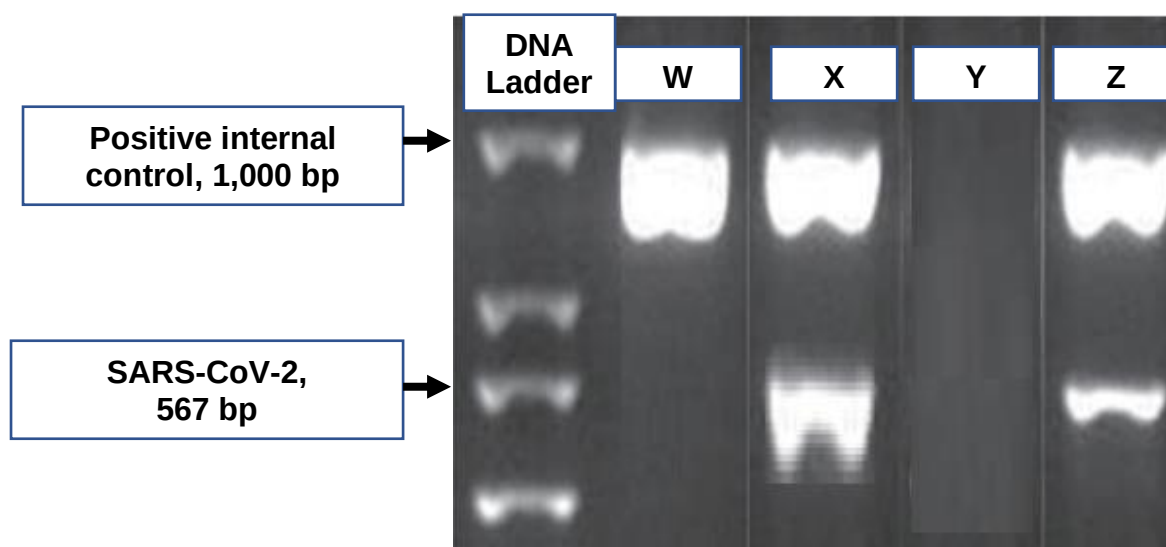


Fig. 8.3

Based on Fig. 8.3, identify which individual/s could be released from quarantine. Explain your answer.

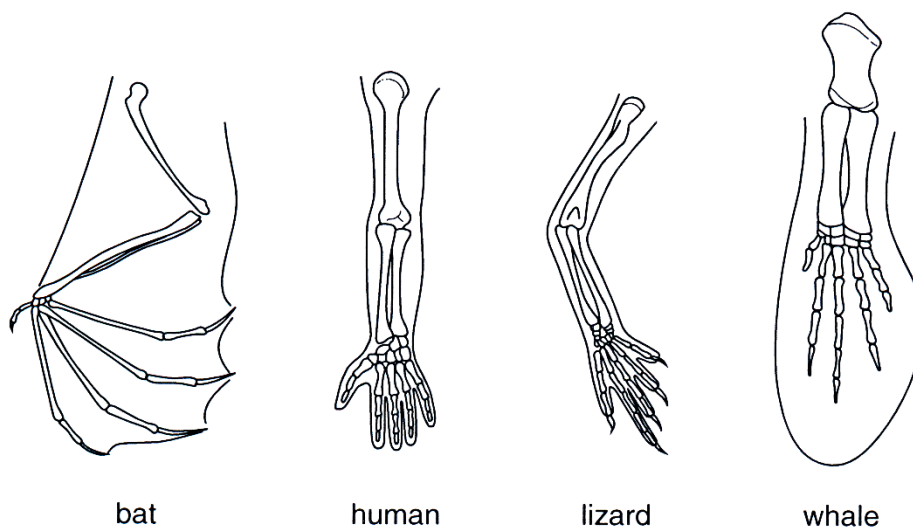
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[2]

[Total: 12]

QUESTION 9

Figure 9.1 shows the arrangement of bones in the pentadactyl forelimbs of four vertebrates. This is used by many scientists to provide evidence for evolution.

**Fig 9.1**

(a) (i) State the term used to describe the relationship between structures such as those in Fig. 9.1.

.....[1]

(ii) Explain how the relationship between the structures in Fig. 9.1 provides evidence to support the theory of evolution.

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[3]

(b) Suggest how natural selection could have brought about the evolution of the skeleton of a bat's wing.

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When comparisons were made with *cytochrome c* gene of the four vertebrates, there was strong evidence to show that these vertebrates share a common ancestry.

(c) (i) Explain how molecular homology could be used to support the claim that these vertebrates share an evolutionary relationship.

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Scientists also make use of fossil records to see the evolution of species through the modification of homologous structures from a common ancestor to the present descendant through a series of transitional forms.

However, the use of these fossils to support Darwin's theory of evolution may be inconclusive.

(ii) Explain why this is so and how molecular data is able to overcome the limitations of the study of fossil records.

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[Total: 11]

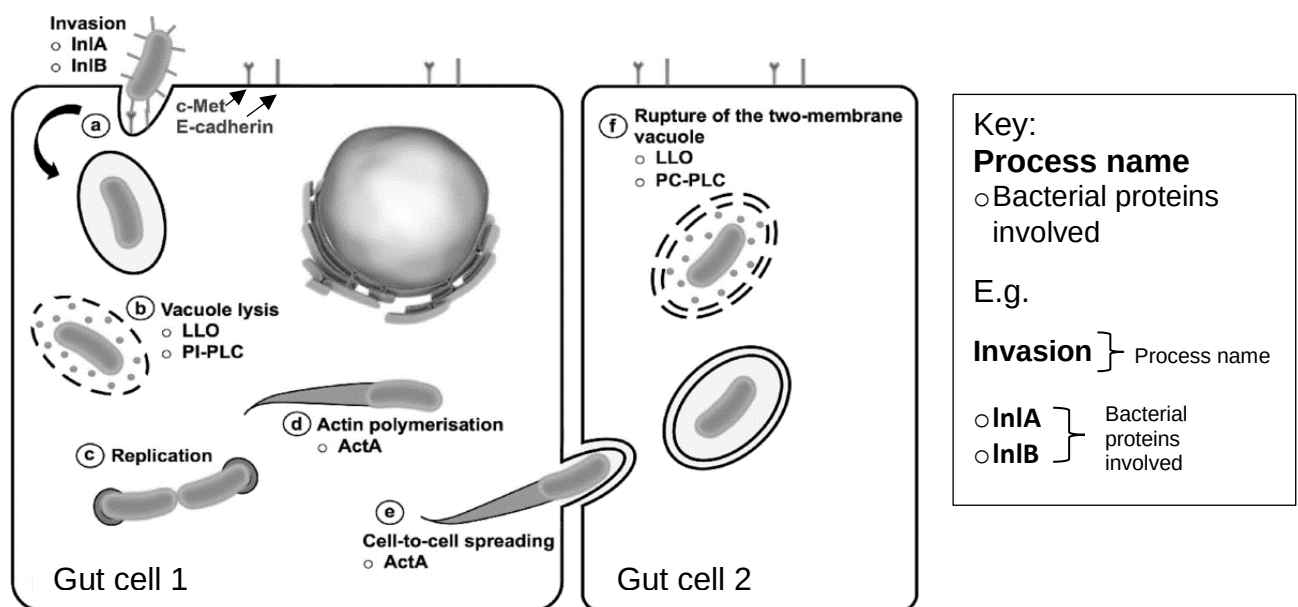
QUESTION 10

Listeria monocytogenes is a type of bacteria that is commonly found in water, soil, and faeces. Humans are infected when they consume foods contaminated with the bacteria.

The bacterium first invades the human host through entering the epithelial cells lining the gut.

It has the capacity to then cross the intestinal barrier and also the blood–brain barrier and the placental barrier, infecting the brain and placenta as well to cause encephalitis (inflammation of the brain), and spontaneous abortions in pregnant women.

Figure 10.1. shows the reproductive cycle of the bacterium in the host cell. This bacterium is known to form a “comet tail” using host cell actin to spread to other host cells (Bonazzi *et. al*, 2009).



Antimicrobial Resistance in *Listeria* Species - Scientific Figure on Research Gate. Available from: https://www.researchgate.net/figure/L-monocytogenes-intracellular-life-cycle-a-Listeria-invades-the-host-cells-via-a_fig2_326528987

Fig. 10.1

(a) Suggest why *L. monocytogenes* can also specifically cause disease symptoms in the brain and induce spontaneous abortions in its human host.

.....[1]

(b) With reference to Fig. 10.1, explain one way in which the invasion by *L. monocytogenes* can affect the gut cells.

.....[1]

(c) With reference to your knowledge of the adaptive immune system, outline how the bacterial-infected gut cells may work together with lymphocytes to respond to the infection.

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[2]

(d) Assuming that a *L. monocytogenes* cell has a spontaneous mutation in one of its chromosomal genes to evade the immune system, list two processes that can facilitate the conferring of this evasion between bacteria cells.

Process 1 Process 2 [1]

(e) Antibiotics such as penicillin are generally prescribed for bacterial infections by *L. monocytogenes*. Explain why penicillin is effective.

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[2]

[Total: 7]

End of Paper